

Confidence Intervals (Part II)

STAT 24510-30040

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5 Wilson's confidence interval (the score method)

The confidence intervals proposed by E. B. Wilson (American mathematician and statistician and more) overcome some problems the Wald-type intervals have. The method is also called the score method, a term we will come across later.

To construct confidence intervals for a population mean estimated by sample mean, Wilson's method directly use the Central Limit Theorem approximation

$$T_n = \frac{\bar{X} - \mathbb{E}(\bar{X})}{\sqrt{\text{Var}(\bar{X})}} \sim N(0, 1) \quad (\text{approximately, for large } n)$$

without employing another approximation that replaces the $\text{Var}(\bar{X})$ (a function of unknown parameters) in the denominator by an estimator, which is the second approximation used in the Wald method.

Wilson's confidence interval for the Poisson mean

Data: $X_1, \dots, X_n$.

Model: X_i 's are *i.i.d.* $\sim \text{Poisson}(\lambda)$.

Goal: Construct a Wilson's confidence interval for λ .

We already know that the MLE $\hat{\lambda} = \bar{X} \sim \text{Poisson}(\lambda)$. By the Central Limit Theorem, when n is sufficiently large,

$$T_n = \frac{\bar{X} - \mathbb{E}(\bar{X})}{\sqrt{\text{Var}(\bar{X})}} = \frac{\hat{\lambda} - \lambda}{\sqrt{\lambda/n}} \sim N(0, 1) \quad (\text{approximately})$$

Use the symmetry of normal density, the asymptotic distribution implies

$$\mathbb{P}\left(z_{\alpha/2} \leq \frac{\hat{\lambda} - \lambda}{\sqrt{\lambda/n}} \leq z_{1-\alpha/2}\right) = \mathbb{P}\left(\left|\frac{\hat{\lambda} - \lambda}{\sqrt{\lambda/n}}\right| \leq z_{1-\alpha/2}\right) \approx 1 - \alpha$$

We want to re-express the inside inequality to get the form

$$\mathbb{P}(L \leq \lambda \leq U) \approx 1 - \alpha$$

Note that

$$\left|\frac{\hat{\lambda} - \lambda}{\sqrt{\lambda/n}}\right| \leq z_{1-\alpha/2} \iff |\hat{\lambda} - \lambda| \leq z_{1-\alpha/2} \sqrt{\frac{\lambda}{n}} \iff (\hat{\lambda} - \lambda)^2 \leq \frac{z_{1-\alpha/2}^2}{n} \lambda$$

Geometric intuition

In the last inequality, the left hand side $(\hat{\lambda} - \lambda)^2$ is a quadratic function of λ , and the right hand side is a linear function of λ with slope $z_{1-\alpha/2}^2/n$, both have the range $\lambda \in (0, \infty)$.

Thus the λ satisfying the inequality must be the λ 's for which the quadratic curve $(\hat{\lambda} - \lambda)^2$ is lower than the line $\frac{z_{1-\alpha/2}^2}{n} \lambda$.

Algebraic derivation

To construct the Wilson's confidence interval with level α , we want to solve the inequality

$$(\hat{\lambda} - \lambda)^2 \leq \frac{z_{1-\alpha/2}^2}{n} \lambda$$

for λ . Moving all the terms to the left and combine terms, we obtain

$$\lambda^2 - \left(2\hat{\lambda} + \frac{1}{n} z_{1-\alpha/2}^2\right) \lambda + \hat{\lambda}^2 \leq 0$$

This is a quadratic inequality of λ of the general form

$$\lambda^2 + b\lambda + c \leq 0$$

The inequality = 0 when λ has the values

$$\frac{-b \pm \sqrt{b^2 - 4c}}{2}$$

These are the two endpoints $[L, U]$ of the desired confidence interval for λ .

Enter the values of b, c , we obtain the Wilson's confidence interval for λ as

$$[L, U] = \left[\hat{\lambda} + \frac{1}{2n} z_{1-\alpha/2}^2 - \frac{\sqrt{\left(2\hat{\lambda} + \frac{1}{n} z_{1-\alpha/2}^2\right)^2 - 4\hat{\lambda}^2}}{2}, \hat{\lambda} + \frac{1}{2n} z_{1-\alpha/2}^2 + \frac{\sqrt{\left(2\hat{\lambda} + \frac{1}{n} z_{1-\alpha/2}^2\right)^2 - 4\hat{\lambda}^2}}{2} \right]$$

the confidence level is $(1 - \alpha)100\%$, because

$$\mathbb{P}(L \leq \lambda \leq U) = \mathbb{P}\left(\left|\frac{\hat{\lambda} - \lambda}{\sqrt{\lambda/n}}\right| \leq z_{1-\alpha/2}\right) = \mathbb{P}\left(z_{\alpha/2} \leq \frac{\hat{\lambda} - \lambda}{\sqrt{\lambda/n}} \leq z_{1-\alpha/2}\right) \approx 1 - \alpha$$

Remarks

- Note that the confidence interval is no longer centered at $\hat{\lambda}$. In fact, the center of the confidence interval is $\hat{\lambda} + \frac{1}{2n} z_{1-\alpha/2}^2$, which is not an unbiased estimator of λ , but the estimator is asymptotically unbiased as $n \rightarrow \infty$.
- As implied by the inequality $(\hat{\lambda} - \lambda)^2 \leq \frac{z_{1-\alpha/2}^2}{n} \lambda$, the confidence interval $[L, U] \subset (0, \infty)$, which is consistent with the range of λ . This is an advantage of the Wilson's confidence interval over Wald type confidence interval.
- For larger n , the curve on the left hand side of $(\hat{\lambda} - \lambda)^2 \leq \frac{z_{1-\alpha/2}^2}{n} \lambda$ stays the same, while the slope of the line on the right hand side becomes smaller for larger n , thus yields closer endpoints, consequently narrower interval. This demonstrates that larger samples produce narrower, more precise confidence intervals.

- For the geometric intuition interpretation, instead of comparing the quadratic curve $(\hat{\lambda} - \lambda)^2$ and the line $\frac{z_{1-\alpha/2}^2}{n} \lambda$, we may stay with the equivalent inequality $|\hat{\lambda} - \lambda| \leq z_{1-\alpha/2} \sqrt{\frac{\lambda}{n}}$ and state that, the λ satisfying the inequality (thus in the C.I.) are the λ 's for which the curve of $|\hat{\lambda} - \lambda|$ is lower than the curve of $\frac{z_{1-\alpha/2}}{\sqrt{n}} \sqrt{\lambda}$. Here the squared version removes $\sqrt{*}$, thus cleaner. The algebraic derivation and results will be the same.

Wilson's confidence interval for the Bernoulli-Binomial parameter

Data: $X_1, \dots, X_n$.

Model: X_i ' are *i.i.d.* $\sim \text{Bernoulli}(p)$.

Goal: Estimate the parameter p and provide a confidence interval for p , using the Wilson's method.

We know the MLE $\hat{p} = \bar{X}$, and $\sum_{i=1}^n X_i = n\hat{p} \sim \text{binomial}(n, p)$. By the Central Limit Theorem,

$$T_n = \frac{\bar{X} - E(\bar{X})}{\sqrt{\text{Var}(\bar{X})}} = \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \sim N(0, 1) \quad \text{asymptotically for large } n.$$

Wilson's confidence interval is constructed from this asymptotic distribution directly.

Again, by the symmetry of normal density, the asymptotic distribution leads to

$$\mathbb{P}\left(z_{\alpha/2} \leq \frac{\hat{p} - p}{\sqrt{p(1-p)/n}} \leq z_{1-\alpha/2}\right) = \mathbb{P}\left(\left|\frac{\hat{p} - p}{\sqrt{p(1-p)/n}}\right| \leq z_{1-\alpha/2}\right) \approx 1 - \alpha$$

In order to re-express the inside inequality to get the form $\mathbb{P}(L \leq p \leq U) \approx 1 - \alpha$, we look into the p 's satisfying the inequality inside the last probability

$$\left|\frac{\hat{p} - p}{\sqrt{p(1-p)/n}}\right| \leq z_{1-\alpha/2}$$

If we want to avoid dealing with square root, we may square both sides of the inequality to have an equivalent inequality.

$$\left|\frac{\hat{p} - p}{\sqrt{p(1-p)/n}}\right| \leq z_{1-\alpha/2} \iff (\hat{p} - p)^2 \leq \frac{z_{1-\alpha/2}^2}{n} p(1-p)$$

Geometric intuition

The left hand side of the last inequality is $(\hat{p} - p)^2$, a quadratic function of p open upward; and the right hand side $\frac{z_{1-\alpha/2}^2}{n} p(1-p)$ is a quadratic function of p open downward, both curves have the domain $p \in [0, 1]$.

Thus the values of p satisfying the inequality are the p 's for which the open-upward quadratic curve $(\hat{p} - p)^2$ is lower than the open-downward quadratic curve $\frac{z_{1-\alpha/2}^2}{n} p(1-p)$.

Algebraic derivation

To construct the Wilson's confidence interval, we need to solve the inequality

$$(\hat{p} - p)^2 \leq \frac{z_{1-\alpha/2}^2}{n} p(1-p)$$

for p . Moving all the terms to the left and combine terms by the power of p , we obtain

$$(n + z_{1-\alpha/2}^2) p^2 - (2n\hat{p} + z_{1-\alpha/2}^2) p + n\hat{p}^2 \leq 0.$$

This is a quadratic inequality of p of the form

$$ap^2 + bp + c \leq 0$$

and the equality holds when p has the values

$$\frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

These two values are the two endpoints $[L, U]$ of the $(1 - \alpha)100\%$ Wilson's confidence interval,

$$[L, U] = \left[\frac{(2n\hat{p} + z_*^2) - \sqrt{(2n\hat{p} + z_*^2)^2 - 4n\hat{p}^2(n + z_*^2)}}{2(n + z_*^2)}, \frac{(2n\hat{p} + z_*^2) + \sqrt{(2n\hat{p} + z_*^2)^2 - 4n\hat{p}^2(n + z_*^2)}}{2(n + z_*^2)} \right]$$

where $z_* = z_{1-\alpha/2}$ denote the quantile of $N(0, 1)$.

The weighted average interpretation

Simplify, the expression of the endpoints of Wilson's confidence interval can be written as

$$\frac{(2n\hat{p} + z_*^2) \pm z_* \sqrt{z_*^2 + 4n\hat{p}(1-\hat{p})}}{2(n + z_*^2)} = \frac{(2n\hat{p} + z_*^2) \pm z_* \sqrt{z_*^2 + 4n\hat{p}(1-\hat{p})}}{2n^*}$$

where

$$n^* = n + z_*^2, \quad z_* = z_{1-\alpha/2}$$

Furthermore, rewrite the endpoints as

$$\hat{p}w + \frac{1}{2}(1-w) \pm z_* \sqrt{\frac{\hat{p}(1-\hat{p})}{n^*}w + \frac{(\frac{1}{2})(1-\frac{1}{2})}{n^*}(1-w)},$$

where the weight

$$w = \frac{n}{n^*} \in (0, 1)$$

This formulation shows that the score (Wilson) confidence interval can be interpreted as an adjusted weighted average of the observed sample proportion $\hat{p}$ and the midpoint $\frac{1}{2}$.

Remarks

- The Wilson's confidence interval is not centered at $\hat{p}$, the unbiased estimator of p .
- The center of the confidence interval, if used as an estimator of p , is biased, but is asymptotically unbiased.
- As implied by the inequality $(\hat{p} - p)^2 \leq \frac{z_{1-\alpha/2}^2}{n} p(1-p)$, the confidence interval $[L, U] \subset [0, 1]$, which is consistent with the range of p .

In this regard, the Wilson's confidence interval is better than Wald's, since it is possible in practice that Wald confidence interval, being symmetric with respect to $\hat{p} = \bar{X}$, may yield endpoints outside of $[0, 1]$.

To amend such cases, a truncation is needed to limit the confidence interval $\subset [0, 1]$.

- For the geometric intuition, as in the Poisson example, instead of comparing the squared version, we may work with the original inequality $\left|\frac{\hat{p} - p}{\sqrt{p(1-p)/n}}\right| \leq z_{1-\alpha/2}$ and state that, the p satisfying the inequality (thus in the C.I.) are the p 's for which the curve of $|\hat{p} - p|$ is lower than the curve of $\frac{z_{1-\alpha/2}}{\sqrt{n}} \sqrt{p(1-p)}$.

Again, the squared inequality $(\hat{p} - p)^2 \leq \frac{z_{1-\alpha/2}^2}{n} p(1-p)$ makes the calculation easier by removing the square root. The algebraic derivation and results stay the same.

Comparison of Wald and Wilson's confidence intervals (for population mean)

- The Wald confidence interval
 - Simple formular. Easy to compute.
 - Most common.
Wald C.I. is the default C.I. in just about all statistics software packages.
 - The confidence interval is symmetric with respect to the estimator (the sample mean).
 - If the range of the parameter is bounded in certain direction, the default bounds of Wald confidence intervals could be out of the range.
Such cases happen with small probability, usually when sample size is relatively small.
If out-of-bound cases could happen, the researcher should adjust the procedure by appropriate truncations.
 - The Wald test statistic uses the sample standard error of the estimator.
For example, in the Bernoulli-binomial case, the population mean is the proportion p , and the sample proportion is $\hat{p}$. In the test statistics T_n , the standard error of $\hat{p}$ in the denominator is estimated at $p = \hat{p}$, the maximum likelihood estimator of p .

- The score (Wilson) confidence interval
 - The coverage probability (i.e., the confidence level) is more accurate than Wald C.I.
 - Wilson's confidence interval is contained within the range of possible values of the parameter in general, therefore it is more reasonable than Wald confidence interval when the parameter has bounded borders.
This property is more prominent when the parameter estimate is near the border of the parameter space.
For example, in the binomial case, when $\hat{p} \approx 0$ or $\hat{p} \approx 1$, especially when the sample size n is small, Wilson's C.I. $\subset [0, 1]$, thus is more sensible.
 - The endpoints are harder to calculate by hand, but not a problem nowadays with handy computing power.
 - The center of Wilson's C.I. is no longer an unbiased estimator of the parameter.
In the binomial example, the center of Wilson's confidence interval is a weighted average of the sample proportion $\hat{p}$ and $\frac{1}{2}$, thus not symmetric with respect to the unbiased estimator $\hat{p}$.
 - the score (Wilson) test statistic used the standard deviation of the estimator directly.
 - The duality property
 - * The score (Wilson) C.I. is a type of duality confidence interval with respect to hypothesis testing.
For example, in the Bernoulli-binomial case, the score confidence interval of p consists of all possible values of p_0 that, at the test level α , the null hypothesis $H_o: p = p_0$ is not rejected based on the observed sample proportion $\hat{p}$.
 - * In the hypothesis testing point of view, the score (Wilson) test statistic used the standard error estimated at $p = p_0$, for each p_0 where the null hypothesis $H_o: p = p_0$ is not rejected.
 - * The following section comment on the duality property.

Wilson's confidence interval as duality of hypothesis testing

Wilson's confidence interval can be constructed from a hypothesis testing setup using a duality method. The Wilson's method in this context is often call the score method.

In the following, the Wilson' confidence interval for the Bernoulli-Binomial parameter is re-constructed using a duality method by way of hypothesis testing,

For a fixed significance level α , we consider hypothesis test

$$\begin{cases} H_o : & p = p_o \\ H_a : & p = p_a \end{cases}$$

on a sample of $X_1, \dots, X_n$, *i.i.d.* $\sim \text{Bernoulli}(p)$. the test statistic, called the score test statistic, is

$$T = \frac{\bar{X} - p_o}{\sqrt{p_o(1 - p_o)/n}}$$

By the Central Limit Theorem,

$$\text{Under } H_o, \quad T = \frac{\bar{X} - p_o}{\sqrt{p_o(1 - p_o)/n}} \sim N(0, 1) \quad \text{asymptotically.}$$

Consider all possible p_0 for which the (score) test statistic T would not reject the null hypothesis $H_o : p = p_o$. All and every such p_o should satisfy

$$|T| = \frac{|\sqrt{n}(\hat{p} - p_o)|}{\sqrt{p_o(1 - p_o)}} < z_*$$

where $z_* = z_{1-\alpha/2}$ denote the quantile of $N(0, 1)$. For example, $z_* = 1.96$ for $\alpha = 0.05$.

By duality, the values of all such p_o form an interval, that is, the $(1 - \alpha)100\%$ (score) confidence interval for p by duality.

Omit the subscript in p_o , square both sides, multiply the denominator on both sides, the inequality becomes

$$n(\hat{p} - p)^2 < z_*^2 p(1 - p)$$

Rearrange, we have

$$(n + z_*^2)p^2 - (2n\hat{p} + z_*^2)p + n\hat{p}^2 < 0.$$

This is a quadratic function in p . Solving p in the equation $(n + z_*^2)p^2 - (2n\hat{p} + z_*^2)p + n\hat{p}^2 = 0$, the two solutions have the expression

$$\frac{(2n\hat{p} + z_*^2) \pm \sqrt{(2n\hat{p} + z_*^2)^2 - 4n\hat{p}^2(n + z_*^2)}}{2(n + z_*^2)}$$

Note that these are the endpoints which yield exactly the same Wilson's confidence interval

$$\left[\frac{(2n\hat{p} + z_*^2) - \sqrt{(2n\hat{p} + z_*^2)^2 - 4n\hat{p}^2(n + z_*^2)}}{2(n + z_*^2)}, \frac{(2n\hat{p} + z_*^2) + \sqrt{(2n\hat{p} + z_*^2)^2 - 4n\hat{p}^2(n + z_*^2)}}{2(n + z_*^2)} \right]$$

as we obtained before.